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Chapter 1

Theoretical Background

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1.1 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [1], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.1.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.1.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c.\ , \tag{1.1}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.2}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\mathrm{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\mathrm{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.3}$$

Averaging over a field period [1]:

$$U_{\mathrm{dip}} = -\frac{1}{2}\langle\mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]I, \tag{1.4}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.
It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\mathrm{dip}}(\mathbf{r}) = -\nabla U_{\mathrm{dip}} = \frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.5}$$

The dipole oscillation absorbs a power

$$P_{\mathrm{abs}} = \langle\dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega\mathrm{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\mathrm{Im}[\alpha]I, \tag{1.6}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{sc}(\mathbf{r}) = \frac{P_{abs}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\mathrm{Im}[\alpha]I(\mathbf{r}). \tag{1.7}$$

1.1.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.8}$$

Using $E(t) = \mathcal{E}_0 \exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0 \exp(-i\omega t)$, equation (1.8) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.9}$$

Using $p = -ex(t) = -ex_0 \exp(-i\omega t)$ and (1.9), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.10}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.11}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\,\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.12}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}|\langle e|\,\hat{d}\,|g\rangle|^2, \tag{1.13}$$

where $\hat{d}$ is the dipole moment operator.

Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.8 (for a rigorous derivation see Chapter V of Steck [2]). Therefore, the classical polarizability given by equation 1.12 represent a well suited approximation for ODTs.

1.1.1.3 Dipole potential and scattering rate

Substituting α form (1.12) in (1.4) and (1.7) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right) I(\mathbf{r}), \tag{1.14}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.15}$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of ~ 3.8 THz thus a trap typically works at few Thz of detuning, while $\Gamma \sim 36$ MHz. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta} I(\mathbf{r}) \tag{1.16}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta} U_{\text{dip}} \tag{1.17}$$

Equations (1.16) and (1.17) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.1.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.1.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}} \cdot \mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j \neq i} \frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.18}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + \hbar(n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.18) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.19}$$

where Δ_{ij} is the detuning of the $i \to j$ transition with respect to the laser field.

Considering a two-level atom first, (1.18) has only one term, using equations (1.13) and (1.19) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm \frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.20}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.20) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.16). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. If a low saturation regime is assumed ($I/\Delta \ll 1$), all the atom can be considered in the ground state, and the light-shifted ground state became the effective potential that govern the motion of the atoms which drives them toward region of minima of ΔE (Figure 1.1).

For a multi-level atom equation (1.18) contains many terms similar to equation (1.20), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. However, each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\hat{d}|e_j\rangle$. Such quantity can be expressed as

$$d_{ij} = c_{ij}||d|| \tag{1.21}$$

where $||d||$ is the reduced matrix element given by (1.13) and c_{ij} are real transition coef-

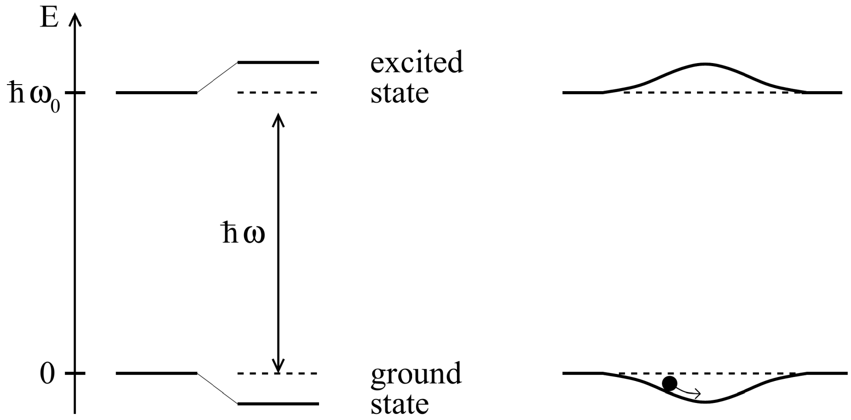

Figure 1.1: *From [1]. Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms.*

ficients that take into account the coupling strength of the two involved states $|g_i\rangle$ and $|e_j\rangle$, and depends on their angular momentum value [1][3].
Therefore, the energy shift (1.18) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.22}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\mathrm{dip}} = \Delta E_i. \tag{1.23}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transition, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.1.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the VIS-IR range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structure are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.
When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is $^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

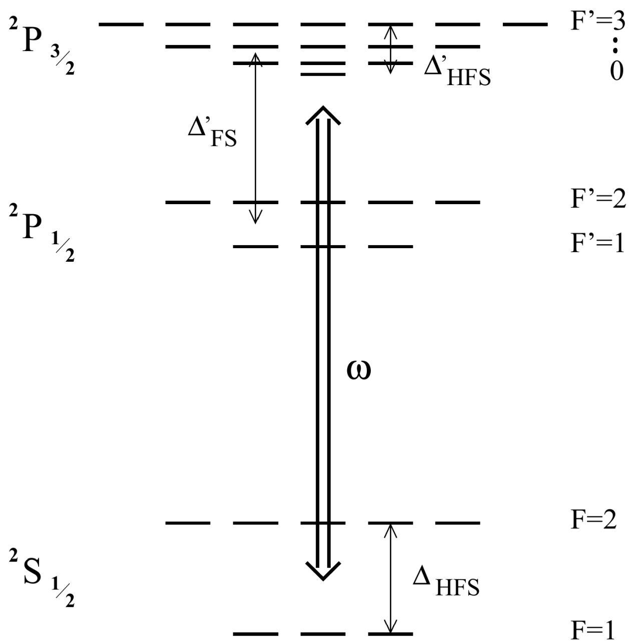

Figure 1.2: *From [1]. Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb.*

- D_1: $^2S_{1/2} \to {}^2P_{1/2}$;
- D_2: $^2S_{1/2} \to {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.2.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.22) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [1]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P} g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P} g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.24}$$

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^\pm$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg$

Δ'_{FS}), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.24) at first order of the small parameter Δ'_{FS}/Δ, one gets [1]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}\left(1+\frac{1}{3}\mathcal{P}g_F m_F \frac{\Delta'_{\text{FS}}}{\Delta}\right) I(\mathbf{r}). \tag{1.25}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.16)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.25) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [1]. Therefore, linearly polarized light is usually preferred for dipole trap implementation.

1.1.3 Heating mechanism

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.1.3.1 Heating source

When atoms interact with laser light, fluctuations in the spontaneous emission and absorption always causes heating (this is also the phenomenon that puts a lower limit to cooling mechanism [4][3]). For the optical dipole trap the atoms absorbs photons and reemits them in random directions (isotropic radiation), statistical oscillations in both these processes cause heating. The absorption of photons from the dipole light is associated

with a recoil energy $E_{\text{rec}} = k_B T_{\text{rec}}/2$ per scattering event and happens only in the direction of the laser beam, hence it is anisotropic. On the other hand, the re-emition scattering events are also associated with a recoil energy E_{rec}, however the emission directions are random, thus the effect is spread over all the three spatial direction. Thus, for a single cycle the atom receives E_{rec} from the absorption and then E_{rec} from the spontaneous emission, for a total of $2E_{\text{rec}}$, in a time $\bar{\Gamma}_{\text{sc}}^{-1}$.
It leads to the total heating power

$$P_{\text{heat}} = 2E_{\text{rec}}\bar{\Gamma}_{\text{sc}} = k_B T_{\text{rec}}\bar{\Gamma}_{\text{sc}}, \tag{1.26}$$

the heating depends on the mean scattering rate $\bar{\Gamma}_{\text{sc}}$ and on the recoil E_{rec} which depends on the laser frequency.
Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [1].

1.1.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy, and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.27}$$

Using equation (1.26) in equation (1.27), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3}T_{\text{rec}}\bar{\Gamma}_{\text{sc}}. \tag{1.28}$$

Expressing the dipole potential as

$$\bar{U}_{\text{dip}} = U_0 + \bar{E}_{\text{pot}} = U_0 + \frac{3}{2}k_B T, \tag{1.29}$$

where U_0 is the potential energy minimum, then the scattering rate $\bar{\Gamma}_{\text{sc}}$ can be expressed by Substituting U_{dip} from (1.17), obtaining

$$\bar{\Gamma}_{\text{sc}} = \frac{\Gamma}{\hbar\Delta}(U_0 + \frac{3}{2}k_B T). \tag{1.30}$$

It is important to notice that, for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, by contrast for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.30) in (1.28), with a depth

$\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\text{red}} = \frac{1}{3} T_{\text{rec}} \frac{\Gamma}{\hbar|\Delta|} \hat{U}, \tag{1.31a}$$

$$\dot{T}_{\text{blue}} = \frac{1}{2} T_{\text{rec}} \frac{\Gamma}{\hbar\Delta} k_B T. \tag{1.31b}$$

Equations (1.31) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

1.2 Optical Fibers

A waveguide is a device that uses dielectric media with high refractive index to confine light within a region of space inhibiting its natural spread typical of free space propagation. The wave that propagates inside a waveguide is called guided mode (or guided wave).

Optical fibers are circularly symmetric waveguide, and are by far the most common waveguide in the world. Following chapter III of Yariv et al. [5], this chapter is devoted to understanding the physical principles that allow optical fibers to guide confined light, with particular focus on the shape of their modes of propagation.

1.2.1 Circularly symmetric wave guides

An optical fiber is model as a cylinder called core with refractive index n_1 surrounded by a hollow cylinder called cladding of refractive index n_2. The refractive index has therefore the step profile

$$n(r) = \begin{cases} n_1, & 0 < r < a \\ n_2, & a < r < b, \end{cases} \tag{1.32}$$

where a is the radius of the core and b is the outer radius of the cladding. Usually the radius of the cladding is much larger that the radius of the core ($b \gg a$), so that the electric field can be considered zero at $r = b$. For example in fiber that guide a single mode of light with $\lambda = 1.5\ \mu$m, the typical core radius is $a = 5\ \mu$m while the cladding external radius is in the order of $b = 100\ \mu$m.

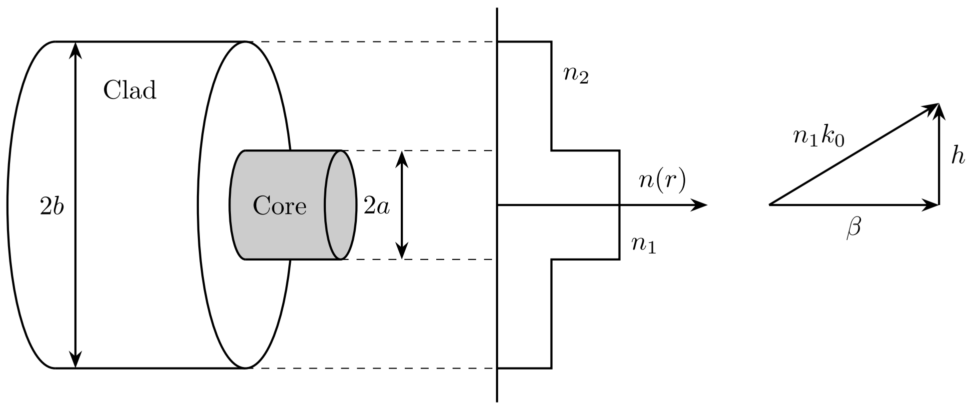

Figure 1.3: *Schematic representation of a step-index optical fiber. (left) and wave vector diagram showing the geometric relationship between the wave number in the core n_1k_0, the longitudinal propagation constant β, and the transverse wave number h (right).*

1.2.1.1 The Cylindrical Wave Equation

Given the symmetry of the system, it is natural to work with the cylindrical coordinates:

$$\begin{cases} \mathbf{u_r} = & \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi \\ \mathbf{u}_\phi = & -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi \\ \mathbf{u_z} = & \hat{\mathbf{z}}, \end{cases} \tag{1.33}$$

thus the electric and magnetic fields are expressed in terms of the components E_r, E_ϕ, E_z, H_r, H_ϕ and H_z. In cylindrical coordinates the unit vectors $\mathbf{u_r}$ and $\mathbf{u}_\phi$ are not constant vectors ($\partial\mathbf{u_r}/\partial\phi = \mathbf{u}_\phi$ and $\partial\mathbf{u}_\phi/\partial\phi = -\mathbf{u_r}$), this complicates the wave equation for the associated components of the field. Nevertheless, as $\mathbf{u_z} = \hat{\mathbf{z}}$ the equation for the z components remain simple:

$$(\nabla_t^2 + k^2)\begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0, \tag{1.34}$$

where $k^2 = \omega^2 n/c^2$ and ∇_t^2 is the transversal component of the Laplacian operator in cylindrical coordinates. To solve the cylindrical wave equation, the usual approach is to solve for E_z and H_z and then express E_r, E_ϕ, H_r and H_ϕ in terms of E_z and H_r, this is possible thanks to Maxwell's curl equations in cylindrical coordinates that allows the r and ϕ components to be expressed as a function of solely E_z and H_z (Appendix A of [5]).

To proceed with the resolution, the dependence on the coordinate z and time t is assumed to be $e^{i(\omega t-\beta z)}$, namely:

$$\begin{bmatrix} \mathbf{E}(\mathbf{r},t) \\ \mathbf{H}(\mathbf{r},t) \end{bmatrix} = \begin{bmatrix} \mathbf{E}(r,\phi) \\ \mathbf{H}(r,\phi) \end{bmatrix} e^{i(\omega t-\beta z)}, \tag{1.35}$$

where β is referred to as the propagation constant of the mode.

Using (1.35) and the transverse cylindrical Laplacian in (1.34), one gets:

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial^2}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\phi^2} + (k^2-\beta^2)\right)\begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.36}$$

Equation (1.36) is solved by assuming the separation of variable

$$F_z(r,\phi) = R(r)\Phi(\phi), \tag{1.37}$$

where $F_z = E_z, H_z$, obtaining

$$\Phi\frac{d^2R}{dr^2} + \frac{1}{r}\Phi\frac{dR}{dr} + \frac{1}{r^2}R\frac{d^2\Phi}{d\phi^2} + (k^2-\beta^2)R(r)\Phi(\phi) = 0, \tag{1.38}$$

which is a separable differential equation.

The variables are separated by dividing by $R\Phi$, obtaining

$$r^2\left(\frac{1}{R}\frac{d^2R}{dr^2}+\frac{1}{r}\frac{1}{R}\frac{dR}{dr}+(k^2-\beta^2)\right)=-\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}. \tag{1.39}$$

The last equation is in the form $f(f)=g(\phi)$, therefore the equality holds $\forall r,\phi$ if and only if the two sides are equal to same constant value:

$$\begin{cases} r^2\left(\frac{1}{R}\frac{d^2R}{dr^2}+\frac{1}{r}\frac{1}{R}\frac{dR}{dr}+(k^2-\beta^2)\right)=l^2 \\ -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}=l^2 \end{cases} \tag{1.40}$$

The angular part has a simple exponential solution:

$$\Phi(\phi)=C_1e^{+il\phi}+C_2e^{-il\phi} \tag{1.41}$$

where C_1 and C_2 are constants. Since $\Phi(\phi)$ must be single valued ($\Phi(0)=\Phi(2n\pi)$) the value of l is constrained to

$$l=0,1,2,... \tag{1.42}$$

The solution can be therefore expressed as:

$$\begin{bmatrix} E_z \\ H_z \end{bmatrix}=R(r)e^{\pm il\phi}$$

where the differential equation for the radial part is

$$\frac{\partial^2R}{\partial r^2}+\frac{1}{r}\frac{\partial R}{\partial r}+\left(k^2-\beta^2-\frac{l^2}{r^2}\right)R=0 \tag{1.43}$$

which is the Bessel differential equation, whose solution is given by a combination of the Bessel functions.

1.2.1.2 The Bessel functions

Equation (1.43) admits different solution depending on the sign of $k^2-\beta^2$:

- if $k^2-\beta^2>0$ the solutions are the Bessel functions of the first and second kind of order l:
$$R(r)=\bar{C}_1J_l(hr)+\bar{C}_2Y_l(hr); \tag{1.44}$$
- if $k^2-\beta^2<0$ the solution are the modified Bessel functions of the first and second kind, of order l:
$$R(r)=\bar{C}_1I_l(qr)+\bar{C}_2K_l(qr), \tag{1.45}$$

where $h^2 = k^2 - \beta^2$ and $q^2 = \beta^2 - k^2$, and $\bar{C}_1$ and $\bar{C}_2$ are constants.
The Bessel and modified Bessel functions are showcased in Figure 1.4. For proceed with the physical solution, their asymptotic forms for large and small argument are needed.
For $x \gg 1, l$:

$$\begin{aligned}
J_l(x) &\rightarrow \left(\frac{2}{\pi x}\right)^{1/2} \cos\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right) \\
Y_l(x) &\rightarrow \left(\frac{2}{\pi x}\right)^{1/2} \sin\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right) \\
I_l(x) &\rightarrow \left(\frac{1}{2\pi x}\right)^{1/2} e^{x} \\
K_l(x) &\rightarrow \left(\frac{\pi}{2x}\right)^{1/2} e^{-x}
\end{aligned} \tag{1.46}$$

For $x \ll 1$:

$$\begin{aligned}
J_l(x) &\rightarrow \frac{1}{l!}\left(\frac{x}{2}\right)^{l} \\
Y_0(x) &\rightarrow \frac{2}{\pi}\left(\ln\frac{x}{2} + 0.5772\ldots\right) \\
Y_l(x) &\rightarrow \frac{(l-1)!}{\pi}\left(\frac{2}{x}\right)^{l} \qquad l = 1, 2, 3, \ldots \\
I_l(x) &\rightarrow \frac{1}{l!}\left(\frac{x}{2}\right)^{l} \\
K_0(x) &\rightarrow -\left(\ln\frac{x}{2} + 0.5772\ldots\right) \\
K_l(x) &\rightarrow \frac{(l-1)!}{2}\left(\frac{2}{x}\right)^{l}
\end{aligned} \tag{1.47}$$

Another important properties is that the derivative of a Bessel function of order n is a combination of Bessel functions of orders $n-1$ and $n+1$. Since the r and ϕ components are expressed as a combination of derivatives of E_z and H_z, E_r E_ϕ, H_r H_ϕ are also a combination of Bessel functions of mixed orders.

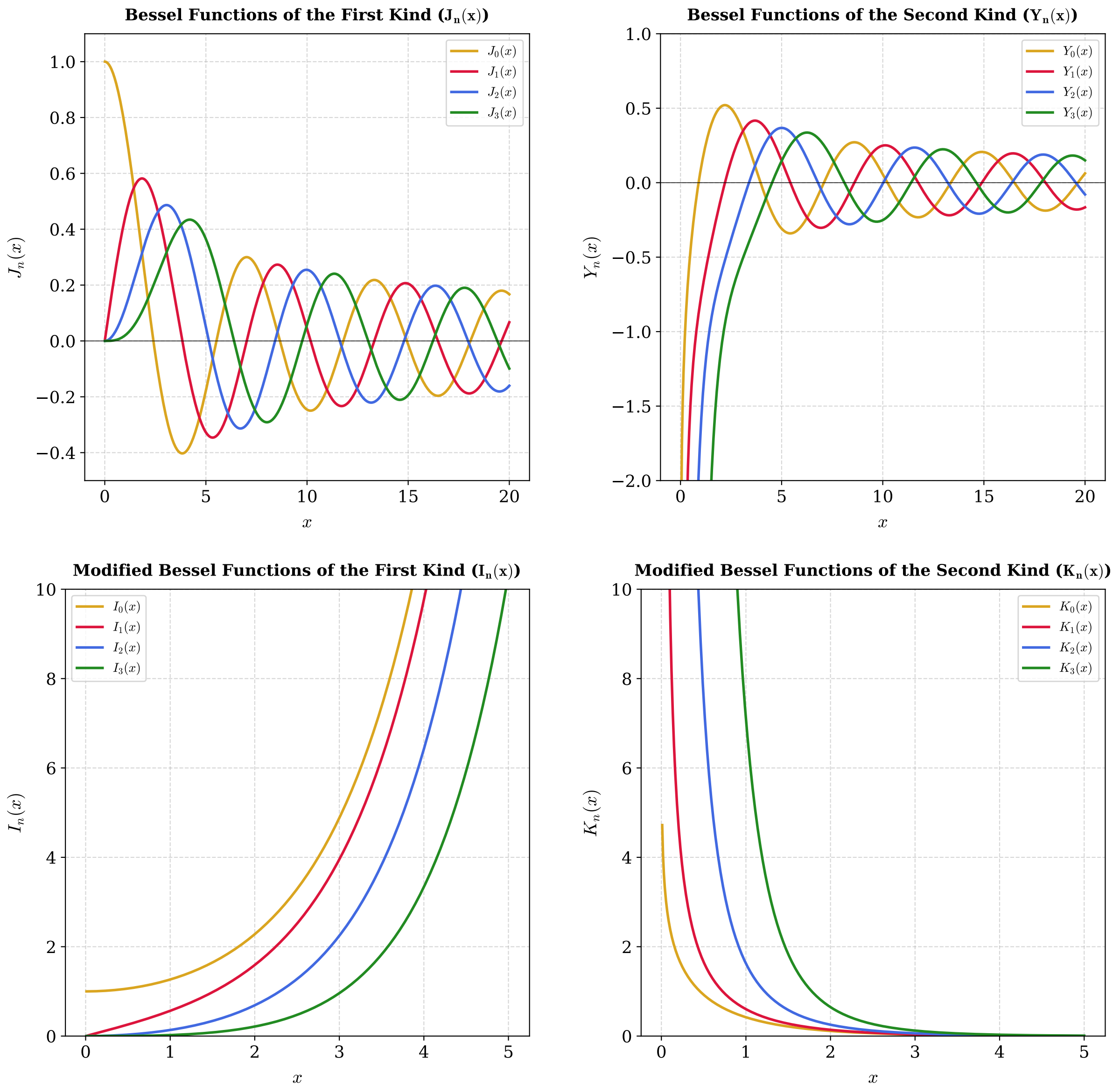

Figure 1.4: *Behavior of the ordinary and modified Bessel functions for the first four integer orders, $n \in \{0, 1, 2, 3\}$. Top panels: Ordinary Bessel functions of the first kind, $J_n(x)$ (left), and second kind, $Y_n(x)$ (right). Bottom panels: Modified Bessel functions of the first kind, $I_n(x)$ (left), and second kind, $K_n(x)$ (right).*

1.2.2 The Weakly Guiding Approximation

The solutions of the complete problem of electromagnetic wave in circular guide are very complicated and lead to solution called hybrid modes indicated by EH and HE, that are a combination of modes with electric field transversal to the propagation of the wave (TE) and modes with the magnetic field transversal to the propagation direction (TM). Hover for a large variety of fibers, the core refractive index is only slightly higher than the refractive index of the cladding:

$$(n_1 - n_2)/n_1 \ll 1, \tag{1.48}$$

this condition which is usually referred to as weak wave guide condition, allows for good approximate LP modes solution, first proposed by Gloge [6].
Using $k^2 = (n(r)k_0)^2$ where $k_0 = \omega/c$, where c is the speed of light, equation (1.36) can be rewritten as

$$[\nabla_t^2 + ((n(r)k_0)^2 - \beta^2)] \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0 \tag{1.49}$$

Looking for a guided mode in the fiber is equivalent to look for a solution that has oscillatory behavior in the core while exhibits vanishing behavior in the cladding. According to common second order equation behavior, the guidance condition introduces the requirements:

$$\begin{array}{lll} r > a & \text{vanishing} & \Longrightarrow (n_2 k_0)^2 - \beta^2 < 0 \\ r < a & \text{oscillating} & \Longrightarrow (n_1 k_0)^2 - \beta^2 > 0 \end{array} \tag{1.50}$$

therefore the propagation constant must satisfy:

$$n_2 k_0 < \beta < n_1 k_0, \tag{1.51}$$

which for the weak waveguide becomes

$$n_2 k_0 \simeq \beta \simeq n_2 k_0. \tag{1.52}$$

The transversal wave numbers h and q are defined as

$$h = \sqrt{n_1^2\, k_0^2 - \beta^2} \tag{1.53}$$

$$q = \sqrt{\beta^2 - n_2^2\, k_0^2}. \tag{1.54}$$

h represents the effective transversal wave number (see Figure 1.3) while q describes the transversal decay rate of the guided mode in the cladding.
According to (1.52), it holds that

$$h, q \ll \beta \tag{1.55}$$

this implies that the rays are approximately parallel to the fiber axis, therefore the longitudinal component of the electric and magnetic field are far weaker than the transverse components and the guided waves are approximately transverse electromagnetic (TEM) [7].

Now the objective is to identify a mode basis that generates all the possible weakly guided modes. The approach is to focus on solutions where either x or y component vanishes, so looking for x-polarized or y-polarized solutions. The approximation made by Gloge is based on postulating for the transverse component a simple dependence on a single Bessel function [6]. Matching the requirements (1.50) with the Bessel equation solution condition (1.44) and (1.45), one deduce that the mode in the core of the fiber must be described by Bessel functions, while in the cladding it has to be described by modified Bessel functions. Furthermore, for $x \to 0$ the mode cannot diverge, and for $x \to \infty$ the mode must vanish, by comparison with the asymptotic behavior (1.46) and (1.47), the mode in the core must be described by a Bessel function of the first kind and the mode in the cladding must be described by a modified Bessel function of the second kind.

Therefore, considering a y-polarized wave, the postulated magnetic field is:

$$\begin{aligned} &E_x = 0 \\ &E_y = \begin{cases} AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r<a \\ BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r>a \end{cases}, \end{aligned} \tag{1.56}$$

where A and B are constant that need to be determined from the continuity condition at the interface of the core and the cladding. It should be noted that for each value of l, this equation describes two distinct modes associated to $+\phi$ and $-\phi$. Taking into account the two possible polarization, this brings the number of basis mode elements to four for each value of l.

Employing the Maxwell's equations, the magnetic field $\mathbf{H}$ can be expressed starting from (1.56) [5]:

$$\begin{cases} H_x = \frac{-i}{\omega\mu}\frac{\partial}{\partial z}E_y = -\frac{\beta}{\omega\mu}E_y \\ H_y \simeq 0 \\ H_z = \frac{i}{\omega\mu}\frac{\partial}{\partial x}E_y. \end{cases} \tag{1.57}$$

And the same holds for E_z:

$$E_z = \frac{i}{\omega\epsilon}\frac{\partial}{\partial y}H_x = \frac{-i\beta}{\omega^2\mu\epsilon}\frac{\partial}{\partial y}E_y. \tag{1.58}$$

Solving the differential equation for E_z and H_z led to the complete form of the basis function in the weak waveguide approximation [5]:

In the core $r < a$:

$$\begin{cases} E_x = 0 \\ E_y = AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \pm\frac{h}{\beta}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} + J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases} \tag{1.59}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{ih}{\omega\mu}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} - J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)}, \end{cases} \tag{1.60}$$

In the cladding $r > a$:

$$\begin{cases} E_x = 0 \\ E_y = BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \pm\frac{q}{\beta}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} - K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases} \tag{1.61}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{iq}{\omega\mu}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} + K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases} \tag{1.62}$$

Considering that $E_z/E_y \propto h/\beta$, $H_z/H_x \propto h/\beta$ in the core and $E_z/E_y \propto q/\beta$, $H_z/H_x \propto q/\beta$ in the cladding, since $h, q \ll \beta$, E_y and H_x are the dominant components, and the mode is approximately TEM, consistently with the considerations done above.
Considering that $E_\phi = -E_x \sin(\phi) + E_y \cos(\phi)$, for a y-polarized mode, $E_\phi \propto E_x$ thus the radial continuity condition at $r = a$ reduces to the continuity condition of E_y, which leads to

$$B = A\frac{J_l(ha)}{K_l(qa)} \tag{1.63}$$

the same reasoning and result holds for H_ϕ continuity.
On the other hand, the continuity condition of E_z at $r = a$ must hold $\forall\phi$, therefore the coefficients of $\exp[\phi(l+1)\phi]$ and $\exp[\pm i(l-1)\phi]$ must be equated separately leading to the two conditions

$$ha\frac{J_{l+1}(ha)}{J_l(ha)} = qa\frac{K_{l+1}(qa)}{K_l(qa)} \tag{1.64}$$

$$ha\frac{J_{l-1}(ha)}{J_l(ha)} = -qa\frac{K_{l-1}(qa)}{K_l(qa)} \tag{1.65}$$

which are equivalent due to the Bessel function properties [5]. The same equations result for the continuity of H_z

The continuity condition of the x-polarized mode lead the same exact requirements.

1.2.3 The LP modes

Equation (1.64) depends only on β through h and q. It has an infinite set of discrete solutions for every l. Hence, the propagation constants solutions of the model are labeled as:

$$\beta_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.66}$$

Their corresponding modes are called Linear Polarized modes, and are labeled as:

$$LP_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.67}$$

Since the condition in (1.64) holds for both x-polarized and y-polarized waves, LP_{lm-x} and LP_{lm-y} are degenerate having the same propagation constant β_{lm}. This can be seen as direct consequence of the circular symmetry of the fiber. Furthermore, considering the degeneracy associated with $\pm\phi$, β_{lm} and LP_{lm} results to be four-fold degenerate, except for the important case $l = 0$ for which the exponential is equal to one independently on the sign of ϕ, resulting in a two-fold degeneration of β_{lm} and LP_{lm}.
In order to determine the allowed values of β_{lm}, the characteristic equation (1.64)must be solved. Since the transverse wave numbers h and q are interdependent, it is convenient to express the in terms of a dimensionless parameter. For this purpose, normalized frequency, also called fiber V parameter, is introduced and defined as:

$$V = k_0 a\sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \sqrt{(ha)^2 + (qa)^2}. \tag{1.68}$$

The V parameter is a fixed parameter of a fiber that depends on its structural parameters, as shown below it determines the number of guided modes and their confinement factor. Using the V parameter, the cladding variable qa can be expressed as

$$qa = \sqrt{V^2 - (ha)^2}, \tag{1.69}$$

and by substituting it back in equation (1.64), the characteristic equation is reduced to a function of the single variable ha, allowing for a graphical or numerical solution. Moreover, equation (1.69) highlights that the left-hand side of (1.64) exists only for $ha < V$.
Figure 1.5, shows the graphical resolution for $l = 0$ (left) and $l = 1$ (right). Considering the case of $l = 0$, the left-hand side of equation (1.64) vanishes at root of $J_1(ha)$, which occurs at $ha = 0,\ 3.832,\ 7.016$ ... While the nominator vanishes at root of $j_0(ha)$ which occurs at $ha = 2.405,\ 5.520,\ 8.654$ corresponding to asymptotes to $\pm\infty$. Thus, in the range $ha \in [0, 2.405]$ the left-hand side is a positive strictly increasing function going from 0 to $+\infty$. The right-hand side, instead, at $ha = 0$ assumes the positive value $VK_1(V)/K_0(V)$

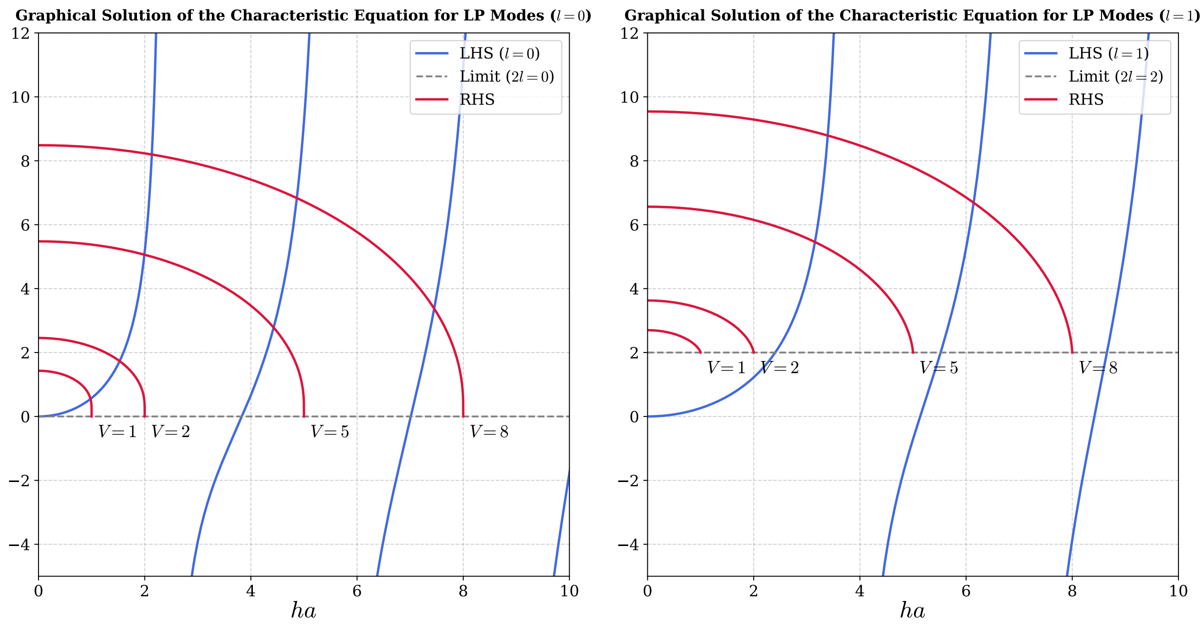

Figure 1.5: *Graphical solution of*

and becomes to decrease as ha increases reaching 0 for $ha = V$. Therefore, there is always an intersection even for very small values of V, meaning that the mode LP_{01} is always guided by an optical fiber regardless of its V parameter.

For $l \geq 1$, the value of the right-hand side in the limit $ha \to V$ can be calculated using (1.47), so that is $2l$. In this situation an intersection with the right-hand side is not guaranteed for every value of V. In particular for $V < 2.405$ the only possible guided mode is the LP_{01}. As V increases beyond 2.405 also the LP_{11} appears, therefore $V = 2.405$ is called the V cutoff value of the LP_{11}.

The V_{cutoff} of a mode can be determined by considering that the mode became guided when the transverse wave vector h associated to that mode satisfies $ha = V$ or equivalently when $q = 0$. Using $q = 0$ in equation (1.65), one gets

$$V_{\text{cutoff}} \frac{J_{l-1}(V_{\text{cutoff}})}{J_l V_{\text{cutoff}}} = 0. \tag{1.70}$$

Thus, the cutoff of the modes LP_{lm} are given by:

$$J_{l-1}(V_{\text{cutoff}}) = 0 \tag{1.71}$$

so that the first root of J_{l-1} is the cutoff of LP_{l1}, the second root is the cutoff of LP_{l2} and so on.

Equation (1.65) is used instead of (1.64) because in the first the right-hand side goes to 0 as $ha \to V$, making the cutoff associated to the roots of Bessel functions of the first kind of order $l - 1$.

V_{cutoff}	$m = 1$	$m = 2$	$m = 3$	$m = 4$
$l = 0$	0	3.832	7.016	10.173
$l = 1$	2.405	5.520	8.654	11.792
$l = 2$	3.832	7.016	10.173	13.323
$l = 3$	5.136	8.417	11.620	14.796
$l = 4$	6.379	9.760	13.017	16.224

Table 1.1: *Cutoff values for different mode numbers.*

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